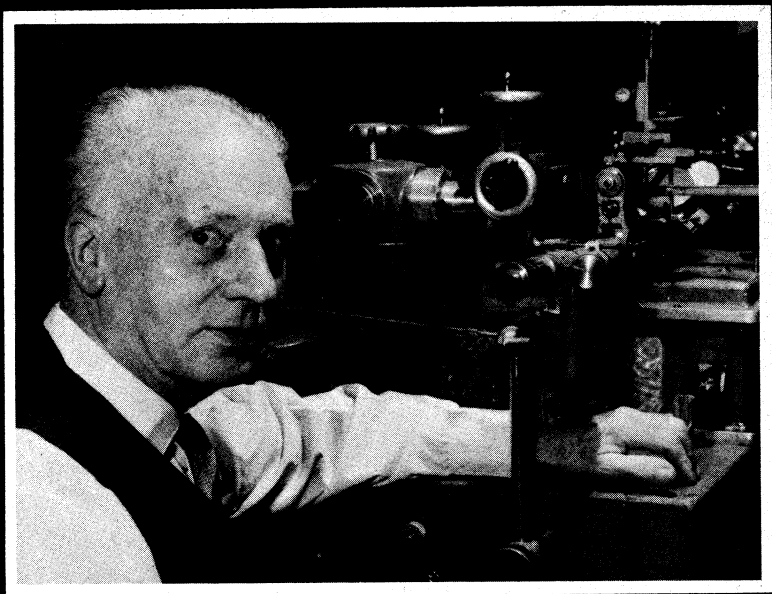


"Professional methods for record care and use"



by Cecil E. Watts, Ltd.
*Internationally Recognized Authority
for Record Maintenance*



In the Tradition of a Perfectionist...

THE research, the methods, the principles and the ingenuity embodied in this text is the work of Cecil E. Watts. The untimely death of Mr. Watts, however, precluded his personal editing of the book.

Mr. Watts was a musician, a recording engineer and a scientist of extraordinary talents. He is regarded by the major recording companies as the definitive authority on record care and maintenance.

Because of the pioneering discoveries he made in this field, Elpa Marketing Industries requested access to Mr. Watts' notes and teachings, and with the invaluable aid of his wife and close associate, B. J. Webb, has compiled this book as the most advanced contribution to the science of preserving disc recordings.

Elpa Marketing Industries proudly dedicates this book to the memory of Cecil E. Watts, and acknowledges with gratitude the gracious help and guidance rendered by Mrs. Watts, his daughter, Susan Watts, and B. J. Webb, without which this publication would not have been possible.



The Era of High Fidelity

WITH the introduction of the long playing record in 1948, high fidelity, as we know it, came of age. Since that time, technological advances have enabled us to come closer to the theoretical ideal of perfect sound reproduction or sound re-creation. As each successive link in the chain of high fidelity reproduction is refined and improved, it necessitates a corresponding improvement in associated components. This has been especially true of the long playing record. We have only to recall the complete revision of the home music system occasioned by the advent of the stereo record to appreciate this relationship.

Since the previous Cecil E. Watts booklet, "How to Clean, Maintain and Protect Records", first appeared in September, 1964, a number of new developments have enabled us to achieve further improvements in the quality of sound reproduction. Perhaps the most significant of these improvements has been in the area of ultra lightweight tracking and the control of static electricity. For our purposes, ultra lightweight tracking refers to stylus pressures of two grams or less. Under these conditions, record wear, (for all practical purposes) is eliminated and a considerable reduction in distortion is achieved, provided the source of sound — the record — is properly maintained and kept free of pollution in its grooves.

Each time we improve one element in the reproductive chain, we must be prepared to make similar improvements in the others. Thus, when we reduce tracking weight to two grams or less, we must of necessity, keep our records more scrupulously clean. All traces of dust, dirt and static electricity must be removed from the record before it can be played. Failure to observe these simple precautions will result in a rapid deterioration of quality and a degradation of the original recorded sound.

When we try to apply old record cleaning techniques and procedures to these newly evolved record playback conditions, we soon discover they are insufficient to the task. Therefore, it has become necessary to re-evaluate old record cleaning procedures, and to devise new ones, to take fullest advantage of improvements in record technology.

If the public can be educated to the necessity of proper record care, then we have gone a long way toward ushering in a new era of enjoyment of recorded music.

...for the record

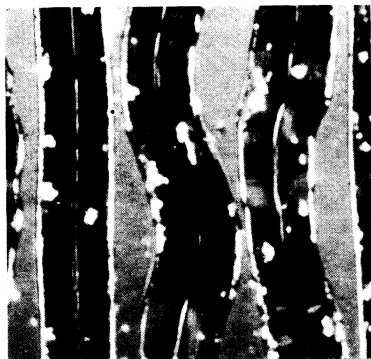
AT this point, a close examination of the interaction between the stylus and the record groove, should give us a better understanding of just how important record cleanliness is.

The musical content of the record is contained within a tiny groove that extends in a tightly wound spiral for more than a half mile. The groove itself is actually narrower than the thickness of a human hair, yet, it is capable of producing an incredible pattern of sounds from the most delicate pianissimo, to a shattering crescendo — from the highest frequencies the human ear can perceive, to fundamentals that are felt rather than heard.

When a record is new and clean, it can produce these variegated sound patterns with astonishing fidelity. But if it is allowed to become dirty and contaminated with pollutants of one kind or another, it loses its magical properties and unless the condition is corrected, it must be discarded.

As for the stylus which is required to trace this delicate groove, it is a marvel of another order. It must be capable of following the thousands of tiny undulations, sharp turns and wide swings (often at incredible velocities), which are contained within the groove and produce a distortion-free replica of the sound. Before stereo, stylus pressures between 4 and 10 grams were the rule. As a result, deformation of the groove with an accompanying deterioration of recorded sound was inevitable. Today, ultra lightweight tracking has virtually eliminated most audible plastic deformation of the record. Consequently, if a record can be maintained in a like-new condition, it will last indefinitely.

To appreciate more fully the interaction between the stylus and the groove, remember that while the pressure exerted on the groove by the stylus is only one or two grams, the contact area is so very small that the pressure developed at any given point is equal to thousands of pounds per square inch.



A new record showing how dust particles are pushed into groove.



Full orchestral recording in good condition.

...in the groove

No discussion of record care would be complete without examining another equally important element in the chain of musical reproduction — the phono pick-up cartridge.

The very small dimensions of the stylus of a modern stereo cartridge, combined with its extreme freedom of movement (high compliance) and its low tracking pressure, require that records be maintained in their original state. Since the pressures are too small for the stylus to push the dirt and dust aside, it “plays” this foreign matter as if it were part of the groove. The result is an increase in background and surface noise.

Even under ideal conditions, as with a new and clean record, the stylus tip will collect some airborne foreign matter. If this deposit is allowed to build up and harden on the stylus, the high polish of the stylus will eventually be replaced by a rough surface which will, in turn, contribute to an increase in surface noise and distortion.

Therefore, we have to keep not only the record clean but the stylus as well. In practice the condition of the stylus is an excellent indicator of the condition of the record. If, upon examination (under a 10X magnifier), the stylus reveals traces of “goo” adhering to the tip, we know that the record has been fouled by contaminants, such as anti-static sprays, grease from fingers, pollution from the air, etc., and the disc must be thoroughly washed following the procedure outlined later in this booklet. On the other hand, if only fluffy particles are found, or the stylus appears clean, the record needs no treatment other than the use of a humid cleaning tool such as the WATTS Preenor or Parastat.



Greasy deposits combine with dust to foul the stylus. Examination of stylus tip after playing a record provides best indication of groove condition. A low power magnifier is adequate for observation.

How to Keep Records Clean

RECORD cleaning is a two-fold problem. Those of you who have read our basic booklet, “How to Clean, Maintain and Protect Records”, know that we must first try to remove every trace of dust, dirt and pollutants from the surface of the record and must also see to it that the cartridge stylus, the turntable mat and the cleaning tools are likewise free of foreign matter. In other words, anything that comes into contact with the record, should be as clean as the record itself.

The second part of the problem is that we must try to eliminate any vestige of a static charge from the record. Once the record becomes electrically charged, it acts like a magnet attracting particles of airborne dust to the surface of the record. All our efforts at keeping the record clean will come to nothing if we cannot prevent or neutralize static charges in the record.

To complicate matters, the very act of cleaning the record can, in itself, induce a static charge. For example, wiping the surface of the record with a cleaning cloth or almost any record cleaning tool can create a static charge. Even playing a record, with the resultant friction between the stylus and the groove, will often induce a static charge.

Making the Method Fit the Equipment

THOUGH we are primarily concerned with lightweight tracking, it is convenient to group various record playing conditions in order of decreasing stylus pressures. The pressure at which a stylus tracks the grooves of the record is a major factor in determining the particular cleaning method necessary. For convenience we can divide them into five groups:

- 1) Stylus pressures between 7 and 10 grams
- 2) Stylus pressures between 5 and 7 grams
- 3) Stylus pressures between 2 and 5 grams
- 4) Stylus pressures below 2 grams
- 5) Elliptical styli.

Groups 1 and 2 pertain to older mono playback equipment, which is inherently incapable of maintaining today's standards of high fidelity reproduction. The records played on this type of equipment inevitably suffer plastic deformation and a consequent degrading of sound quality, but with their limited frequency response this usually goes unnoticed. Record cleaning cloths or sprays should be able to keep these records reasonably clean, since the stylus itself has the ability to brush aside any dirt or dust particles which remain in the grooves. However, even at these higher tracking weights, the records and performance would benefit from the conscientious use of the more advanced record cleaning tools, such as the Dust Bug or the Preener.

As for groups 3, 4, and 5, they are the specific areas in which advanced record cleaning and care are an absolute necessity.

Under ideal conditions, of course, when records are stored in a closed cabinet, and where the air is relatively free from pollution, cleaning problems are less demanding. Such environments are usually found in country residential areas and the people who live in such places are often puzzled by all the fuss about static charges and record cleaning.

However, it is becoming increasingly difficult to escape a polluted atmosphere, so that there are no clearly defined boundaries between areas where records do or do not perform well.

Under these circumstances and with ultra-light tracking conditions, we must concern ourselves with the presence of microparticles of extraneous matter on the record surface. These microparticles which can be seen only with a microscope, are now what we must deal with. Even new records are subject to this problem.

An essential factor in any method used to maintain a record collection in good condition is the attention given to the cleaner itself. Whether you use a Watts Dust Bug, Preenor or one of the Parastats or any other cleaning tool, it must be kept immaculately clean. *The record can only be as clean as the cleaning device.*

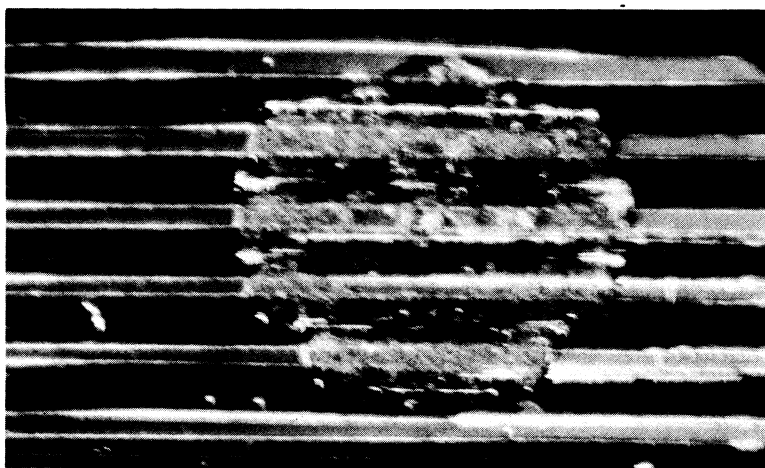
If the dust, dirt and other pollutants which accumulate on the record cleaner are not removed, they will contaminate every record with which the cleaner comes in contact.

This may seem to be an elementary precaution, but it's surprising how many knowledgeable enthusiasts forget this vital detail in their haste to listen to a new record.

Nor should you forget to examine and clean the stylus, which can also accumulate this same extraneous foreign matter. But more on this subject later.

Why is Semi-Dry Cleaning Necessary?

MANY people consider the use of fluids necessary for thorough record cleaning, but the reverse is actually true. The ultra-lightweight tracking of the more sophisticated pick-up cartridges demands that



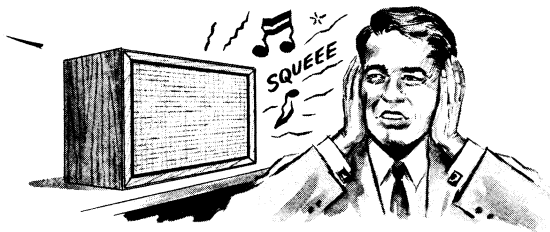
Magnified view shows fine spray of a single droplet of water covering grooves of a record which impedes tracking of stylus.

Why is Semi-Dry Cleaning Necessary? *(continued)*

no free fluid is applied for cleaning or static control, since the stylus can no longer push aside or ride through the wet or hardened liquids to make intimate contact with the grooves of the record.

This necessitates cleaning records in a dry or semi-dry state. Any cleaning device which leaves a film on the record should not be used in groups 3, 4, and 5. In turn, this makes the task of removing the fine particles of dust and dirt from the grooves of the records considerably more difficult.

Effective cleaning tools, therefore, should be in a moist or humid state, but should not be so wet as to leave a film on the surface of the record.



What is Static?

BECAUSE it plays such a major role in the problem of record cleaning, we have repeatedly referred to the pernicious effects of static electricity. Therefore, an explanation of this phenomenon is in order.

Stated simply, static electricity is the charge which results from the friction between one dielectric (non-conductor) and another. The purer the material of which this dielectric is composed, the longer it will hold its static charge.

A static charge can be induced in a variety of ways. For example, it can occur during the pressing process when the record is being made. It can occur if we try to clean the record with a dry cloth. Even removing a record from its sleeve is enough to induce a static charge under certain conditions and it can also result from simply playing the record. Once the static charge is formed, it acts like a magnet and will attract airborne dust and dirt, as well as small solid particles from anything within the immediate vicinity of the record.

Static charges occur when the plastic material of the record becomes negatively charged, that is, when it gains electrons as a result of the friction between it and the cleaning cloth, stylus or record jacket. Since the vinyl material from which the records are made is a dielectric or non-conductor, it will tend to hold and maintain this charge and thus attract dust and dirt.

Shellac records (old 78's) and early L.P.'s were composed of a variety of materials (many of which were impure). As a consequence, the problem of static electricity was of little or no importance, since

these records did not hold a static charge. However, in recent years record companies have gone over to pure vinyl materials in an attempt to increase quality and reduce surface noise. Unfortunately, these pure materials are now much more prone to static charges and hold a static charge longer. Static has now become a major problem.

Older record cleaning techniques overcame the static problem by applying various anti-static fluids to the surface of the record, usually by means of a sponge, spray or impregnated cloth. They had, and still have, their uses with higher tracking forces (5 grams or more). The substances so applied were often viscous, such as grease or gel, and they performed a useful function, in that a film was produced which limited the external field of the static charge and therefore reduced its power to attract airborne dust and dirt. It did not, however, discharge this static.

An added advantage of these anti-static films, was that they acted as a lubricant between the grooves of the record and the stylus, thus reducing record wear with the higher tracking forces being used.

However, with lightweight tracking forces, the stylus is no longer able to go through this film or push it aside but rather rides over it resulting in improper tracking with distorted reproduction and a new kind of surface noise. In addition, an anti-static film will, with repeated applications, result in a sticky or smeary condition which produces a holding or bonding action within the grooves of the record making subsequent removal of dirt and dust much more difficult.

The ultra-lightweight tracking of today's high compliance pick-up cartridges with their greatly reduced tip and moving mass, can only be accomplished if the grooves of the record are almost antiseptically clean, uncontaminated even by microdust or pollution of any kind or by any film or deposit of anti-static substance. Once we understand these factors, we are in a much better position to do something about it.

The obvious solution is to devise a method to allow us to restore and/or maintain our records in a non-charged or quiescent state. If we can get rid of this charge or prevent it from occurring, we will have gone a long way toward solving our record cleaning problems.

Our findings become particularly significant when we also realize that the ability of records to acquire and hold static charges varies according to the humidity in the atmosphere.

A relatively high humidity tends to inhibit the formation of static electricity. In general, no cleaning device should ever be applied to a record in a dry state or static will be automatically generated.

With the single exception of the washing process for contaminated records, any fluid applied to the record surface is potentially harmful, and if any sign of wetness appears on the surface of the record under treatment, the cleaning device is too moist and should not be used in that condition.

We must adapt our record cleaning procedures to suit individual circumstances, i.e., relative humidity, record playing equipment and local atmospheric conditions. What will suffice under one set of circumstances might be inadequate in another.



Elimination of the Static Problem

FORTUNATELY, as far as static is concerned, there are certain measures which can be taken to insure the maintenance of proper playing conditions. Broadly speaking, these countermeasures fall into three categories:

- 1) The use of anti-static films on the surface of the record to control or limit the external attractive field of static charge
- 2) Realignment of the static charge
- 3) Prevention of the static charge by controlling the atmospheric humidity in the immediate vicinity of the record.

In the first instance, i.e., the use of anti-static films, we do not actually remove the static charge, but simply lessen its effect. Moreover, as we have seen, this method has some inherent disadvantages.

In the second procedure, realignment of the static charge away from the playing surface of the record, is accomplished by that much touted device known as the anti-static mat. A great many claims have been made for it. Indeed it has been said that it provides an almost ideal method for eliminating static charges. Actually, this is not the case at all. In practice, anti-static mats simply orient the charge toward the under-side of the record and do not remove, eliminate or make the charge inert. The anti-static mat merely acts like a keeper or a magnet. As a result the realigned static charge attracts the dirt on the mat into the grooves of the record.

As for the third procedure, (prevention of the static charge through humidity control) this is the most desirable method. By creating a humid atmosphere in the *immediate* vicinity of the record we are able to render the record inert. This non-charged condition is what must be maintained if we wish to play our records under ultra-lightweight tracking conditions.

Fortunately, as records and pickup cartridges improved, our knowledge of static and how to deal with it was also being developed at the same time. The result has been the production of anti-static fluids whose viscosity has been greatly reduced. Almost all anti-statics are water soluble, but as the plastic materials on which records are being made become purer and purer they also become highly water resistant.

Through the use of wetting agents we can reduce the viscosity of anti-static solutions so they spread more easily on the record surface. They can now be used at extremely high dilutions without losing efficiency. Under these conditions, no sticky deposit remains in the grooves of the record to combine chemically with the atmospheric pollution and micro-dust to hinder the pickup stylus. In fact, nothing whatsoever is left in the groove after evaporation of the moisture save a molecular trace of anti-static.

Research has also created improved appliances which thoroughly explore the grooves of the record, leaving them perfectly clean, while at the same time applying the necessary trace of anti-static agent and moisture. It has also provided a means of cleaning and restoring discs previously treated with older types of anti-static substance. The results obtained with these new anti-statics, tools and techniques make a number of things clear. The salient points are:

1) Separate tools, possibly of different types, must be used to treat new records, restore older ones and to avoid transfer of dirt.

2) Any cleaning device used, even on a new disc, must itself be thoroughly cleaned immediately after each application to a record surface; otherwise the very pollutants we are trying to remove will merely be spread from one record to another.

3) Static *can* be controlled under any conditions, but will be much more troublesome in some places than others, i.e. centrally heated apartments in cities. Since these are also the environments in which there is a maximum atmospheric pollution, it is under such circumstances that the user of a lightweight pick-up will find it necessary to take the greatest care in record maintenance.

4) The whole key to the control of static lies in the establishment of the correct degree of humidity at the working surface at the time of use. When people store and play their records, under conditions of normal humidity — 75/80% — static charges, once relaxed, will not readily or rapidly build up again, so that complete control is made much easier.

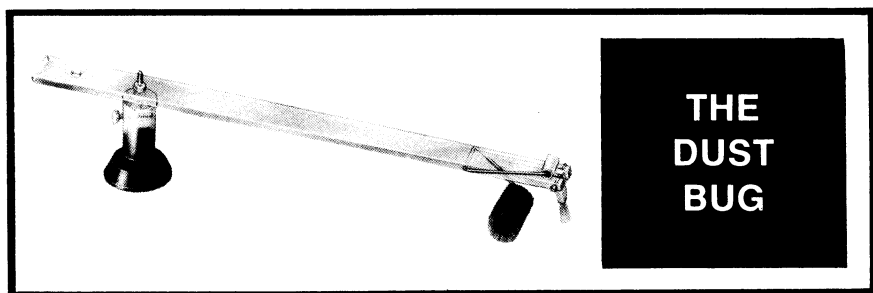
5) Except when a disc has to be washed, no free fluid of any kind should be applied to the disc surface. It would merely dissolve and spread any contamination present. Record manufacturers' instructions still appear on the inner sleeves of their discs to the effect that records should be cleaned with a damp cloth, and as we pointed out in our previous booklet this remains an excellent method of temporarily relaxing a static charge and removing surface dust. It will *not*, of course, explore and remove dust from the groove, and the cloth **MUST** be clean, soft and *barely damp*. *If any sign of wetness, or any mark, appears on the disc when such a cloth, or cleaning device, is used on a disc, then it is too moist, and should not be used in that condition.*

6) The state of the stylus, (as seen when inspected through a magnifier of about X 10 power) provides a clear and immediate indication of the state of the groove.

Cleaning Methods

Now that we know the problems involved in cleaning records, we must ask ourselves the obvious question. What tools are available to accomplish the task?

Over the last ten years, Cecil E. Watts Limited has developed a variety of techniques and tools based on extensive research into the reproduction of quality sound from records. While these tools are not the only ones available, they represent the most outstandingly effective means at our disposal for accomplishing the desired objectives under a wide variety of conditions and circumstances.



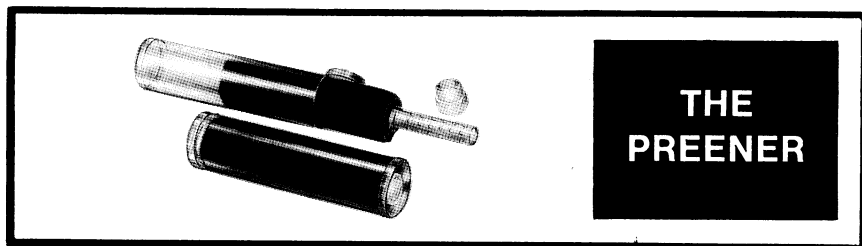
THIS original cleaning tool was developed in the early days of L.P. records when tracking weights of 5 grams or more were common. Its basic purpose was to clean a record and if static was encountered, anti-static fluids or sprays were used in conjunction with it.

With lower tracking weights and increased static problems, a new use has evolved for the Dust Bug. It is ideal for keeping a *clean record* clean while playing, as it will remove the dirt or dust which may settle on the record while it is playing.

Of new importance is the control of static with the ultra-lightweight pick-up cartridges. When the red plush pad is slightly dampened with the anti-static "New Formula" fluid, the pad not only collects all the dirt and dust swept out of the grooves by the nylon brush, but it also supplies the right degree of humidity to prevent the generation of static by friction between the Dust Bug, the stylus and the record. It also relaxes any static charge already present.

The Dust Bug is the simplest of devices, requiring no screw holes or permanent fixing. It can be attached to the turntable instantly and requires little attention except the removal of collected dust after each record is played and application of anti-static fluid to provide the proper trace of moisture or dampness to the plush pad.

Before the Dust Bug is put into use, the record should be cleaned by more sophisticated cleaning devices if ultra-lightweight cartridges are used.

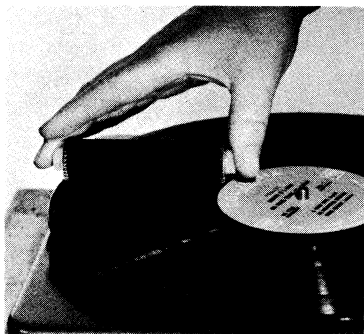


THE Preener is expressly designed for use with new records, or records in new condition. It is able to collect the loose dust and dirt from the record surface without the transfer of any free liquids or fluids. It is used in a moist or dampened condition. It will not, however, remove greasy residues or anti-static films which may have been previously applied.

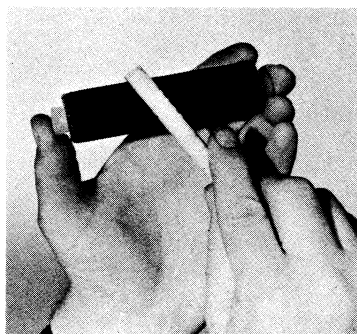
If not judiciously used, it is possible that the micro-dust particles which must be removed for groups 4 and 5, will also not be completely removed by the use of a Preener. To accomplish this objective more easily, the Manual Parastat or the Hi Fi Parastat may have to be employed.

The Preener consists of a roller made of special nylon plush pile impregnated with a non-ionic compound. A wick which is dampened with water and one or two drops of the New Formula anti-static fluid is contained inside the Preener and maintains a damp, moist condition, greatly increasing its efficiency and creating the proper humid condition necessary for the control of static electricity.

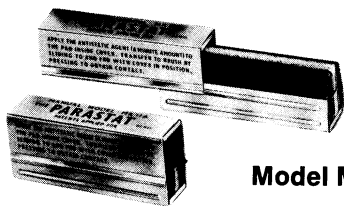
This handy device cleans and preserves the grooves of a new record leaving no trace of film or residue to clog the stylus. To clean the Preener after cleaning the record, remove the accumulated dust from the Preener's surface with the finger for playing conditions in groups 1 and 2 (10 grams to 5 grams tracking weight). For groups 3, 4, and 5, a more thorough cleaning is accomplished by use of the Preener's wick or even better, with the wick of the Humid Mop.



"Preener" in use.



The improved wick of the "Preener" is used to clean its velvet surface and supply the precise humidity required at the moment of use.

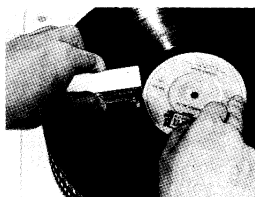


Model MK11A

THE MANUAL PARASTAT

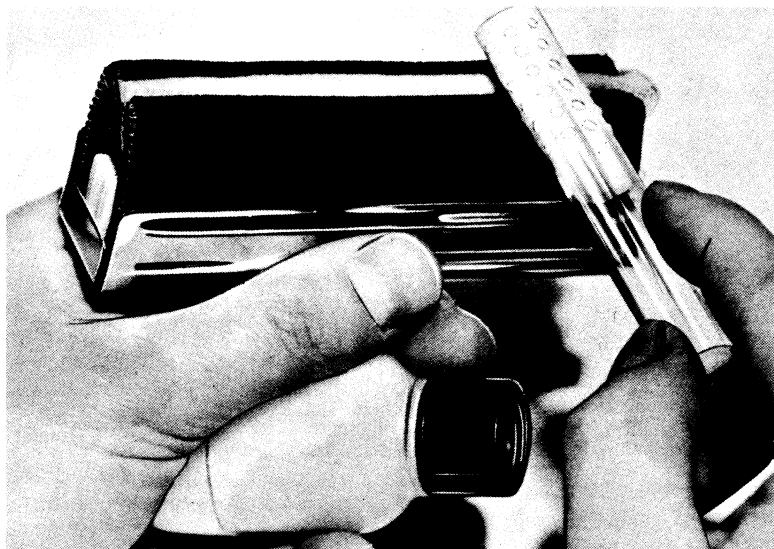
THE Manual Parastat was specifically designed for use with older records which have been treated with anti-static film or which have an excessive accumulation of dust, dirt or residue in their grooves. By design, the Manual Parastat has a dual function. It may be used as a Preener if tilted to either side, since each side is designed like the Preener. As such it may be used for maintenance of new records. It may also be used as a cleaner for old records when held in an upright position so that the nylon bristles between the "two Preeners" can penetrate the record grooves to loosen and remove the accumulated dirt and grit. The brush is made of pure nylon bristles with an average tip radius of 0.00025".

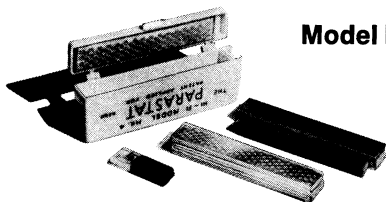
The cover of the Manual Parastat contains an absorbent pad which is brought into contact with the working surfaces by slight pressure during a sliding "to and fro" motion which effectively cleans and moistens the working surfaces. For best results the Manual Parastat should be cleaned using the Humid Mop and its wick.



Manual "Parastat" in use.

The mop is used to remove all collected matter from the "Parastat" surfaces just before use.





Model MK4

THE HI FI PARASTAT

DUE to the very stringent demands imposed by a combination of very low tracking pressures and/or elliptical styli, a new advanced cleaning tool and associated accessories had to be developed for groups 4 and 5. This is the Hi Fi Parastat.

Its intended use is for new records only or for those in new condition. Its object is to clean those records more thoroughly and completely, especially in the sharp corners and minute high frequency configurations of the record groove. At the same time it relaxes any static charge present by simply establishing the correct degree of humidity at the working surface of the record at the time of cleaning.

The Hi Fi Parastat incorporates a humidifier within its case so that it is stored in the proper humid condition. The bristles of the brush have been reduced in thickness as compared to its predecessor, the Manual Parastat, to enable a much greater number of bristles to be used, so that the correct resilience is maintained with increased efficiency. These bristles are now pointed down to almost infinity at an optimum angle insuring complete exploration of the grooves.

To avoid the possibility of any deposit, no matter how minute, in the grooves of the record, only distilled water (supplied with the Hi Fi Parastat) is used to moisten the sponge pad and to dilute the New Formula anti-static fluid which is used when necessary.

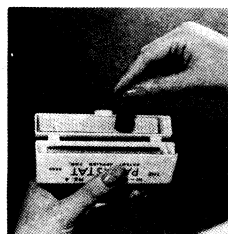
The Hi Fi Parastat is supplied with two dispensers, one containing the New Formula anti-static fluid, the other a drip-feed dispenser of distilled water. In addition, the newly developed WATTS Stylus Cleaner is also supplied. Practical tests have shown that the use of the Hi Fi Parastat and its associated Stylus Cleaner enables the record and the stylus to be maintained in perfect condition for use with the finest available modern equipment.



Applying the anti-static fluid to the humidifier cleaning pad.

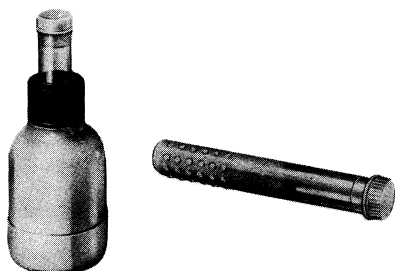


Cleaning the Hi-Fi Parastat brush.



Cleaning the stylus brush.

THE HUMID MOP

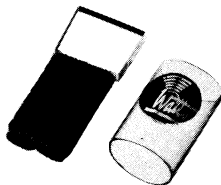


WITH ultra lightweight cartridges (tracking under 2 grams) or with elliptical styli, the surface of the groove walls of the records must be maintained in a perfectly dust free and dry condition. The foreign matter collected by the cleaning tool (Parastat, Preener or Dust Bug) should be removed from the bristles and the velvet pad to prevent it from being returned to the record.

These cleaning tools can be cleaned by rubbing the wick of the Humid Mop, using firm strokes in all directions over the bristles and the velour pads of the cleaning tool being used. This operation rapidly conditions and cleans any of the WATTS cleaning tools and should be used each time the tool is used on a record, thereby assuring the cleanliness of the cleaning tool at all times.

All parts of the Humid Mop are easily separated so that a thorough washing can be accomplished, thereby producing "a one-way street" for accumulated foreign matter.

THE STYLUS CLEANER



THE new WATTS Stylus Cleaner has, as its active cleaning agent, a very close pile nylon pad which will not shed on the stylus tip, yet has sufficient "body" to make it a more efficient tool than a normal brush. It will, however, not loosen or remove a hardened deposit or remove excess "goo" built up over continued tracing of contaminated records. Its purpose and function is to remove all pollution from the stylus which accumulates in a short period of time.

When using the WATTS Stylus Cleaner, the cleaning motion should be from back to front only. In the case of an elliptical stylus the cleaning motion should be at a 45 degree angle, from the back to front, on both sides of the stylus.



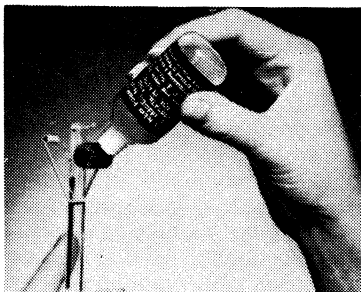
NEW FORMULA ANTI-STATIC FLUID

THIS new type of anti-static fluid was developed early in 1965. It is a completely new departure in the treatment of statically charged records.

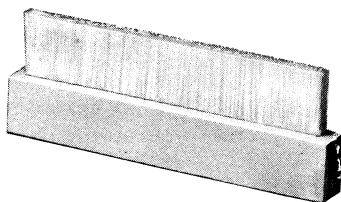
The New Formula fluid contains a highly sophisticated wetting agent developed specially for records. The fluid provides ionic traces of moisture which introduce the required degree of humidity and which reduce the static charge. This is accomplished without requiring a surface film, thereby allowing a "dry state" playback condition without any interfering viscous film.



*2 or 3 drops (not more)
are added to the clean-
ing wick.*



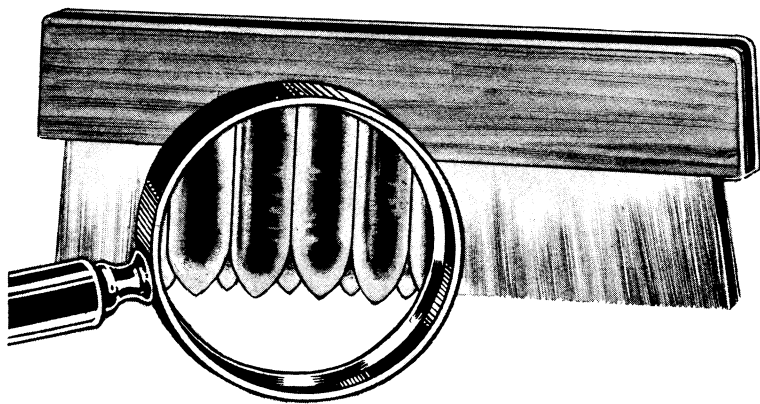
For normal use with 'Dust Bug'



THE RECORD WASH BRUSH

THE Record Wash Brush was developed to penetrate to the very bottom of the grooves of the record. This enables it to remove completely the smallest accumulation of dirt, dust and "goo" embedded in the record. The bristles of this brush are made of pure nylon with an average tip radius of 0.00025".

The brush is used in washing records to remove any previous anti-static films or heavy accumulations of dust and dirt from the grooves. This process is described in another section of this booklet.

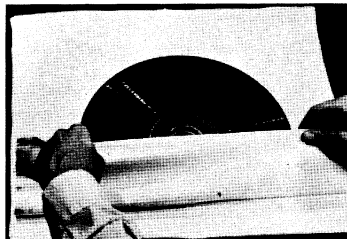


Washing Records

WE have mentioned those circumstances where a thorough washing is the only remedy available. The most difficult situation in record maintenance is a record that has been repeatedly treated with impregnated cloths or sprays with the result that substances or particles have become firmly bonded to the record surface. This is particularly true of the bottom of the groove. As a result, we begin to notice a pronounced distortion of the high frequencies. The only thing we can do now is to wash the record.



Washing a record using a Record Washer Brush to reach bottom of grooves.



Drying the record before final "Parastat" treatment.

To remove all traces of pollution from the record and prevent the fouling of our cleaning devices, we should proceed as follows: First, prepare a clean, lint free drying cloth by spreading it out (without wrinkles) on a clean table. Then prepare a washing solution by using two drops of detergent (a photographic wetting agent is still better) to each pint of clean, warm (90° to 100°F) water. If the local water is very hard, distilled water is preferable. Three to four pints of this mixture in a shallow basin is usually enough. Stand the record in the basin, leaning it on its edge and support it with your thumb on the rim and with

your fingers touching the label area. Wash the record, using a clean, thoroughly rinsed cellulose sponge, or preferably, the WATTS Record Wash Brush, by swilling the solution over its surface. The brush will allow you to reach the very bottom of the groove. Perform this operation in both directions, a section at a time. Rotate the record until the entire surface has been washed several times. Always clean in the direction of the grooves, never across. Repeat the procedure for the reverse side of the record (avoid rubbing the label while it is wet). Then shake off as much excess water as possible and lay the record on one end of the previously prepared drying cloth. Fold the other end over the record and dab gently, until visibly dry.

Finally, place the cleaned record on the turntable and complete the drying operation with a clean Parastat/Humid Mop treatment on both sides. If these directions have been carefully followed, no water will have dried on the record or in the groove. After the record has been played a few times, it should be as good as new.

NOTE: The inner record sleeve should be replaced with a new one, if any anti-static film was present on the record surface before washing.

How to Handle Records

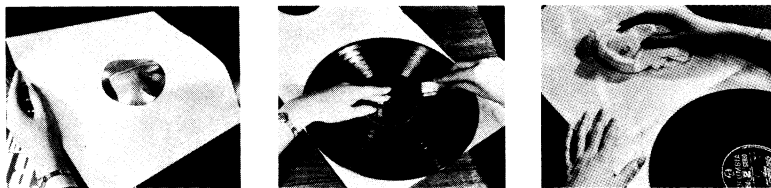
It seems probable that most of the abuse which LP and/or Stereo records are subjected to, stems from the original claims made for them. At the time of their introduction, the new records were hailed as the "ne plus ultra" of technological development. They were said to be flexible, unbreakable and virtually immune to damage. This is not so. Not only are they subject to a variety of "ills" the old shellac records did not suffer from, but it is very difficult to handle them without causing some degradation to the almost invisible recorded grooves which cover most of their surface.

More records are discarded because of damage in handling than are worn out during the actual playing process. Records must always be handled without touching the surfaces (the recorded area) to avoid the transfer of oils from the skin to the record. These oils will act as a bonding element for the accumulation of dust and dirt. They will also result in the formation of unwanted surface films which may cause the stylus to track improperly.

A record should always be removed from its sleeve by inserting the hand until the record can be supported by the fingertips on the underside of the label. Allow the edge of the record to rest in the crook between thumb and forefinger. Then tilt the sleeve until the record slides out. Balance the record in this position until the other hand is free to support the outer edge. It may take some time to acquire the skill to handle records in this manner. If you experience any difficulties, use a clean piece of silk or other soft lintless cloth to prevent finger-marks while holding the record.

Because of difficulties in handling and the possibility of inducing a static charge while removing the record from its sleeve, a new method or technique has been evolved to avoid both of these problems.

If the inner record sleeve is slit on both sides, so that it becomes a folder, we have a ready and easy means for handling the record. Once we remove the sleeve from the record jacket, we need merely open it as we would a book and the record is available to us, without fuss or bother. In addition, we avoid the possibility of a static charge being formed when we try to slip the record out of the sleeve.



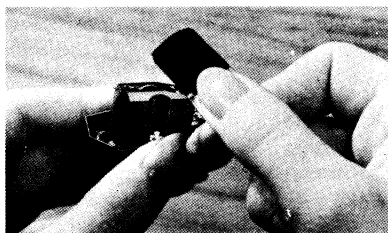
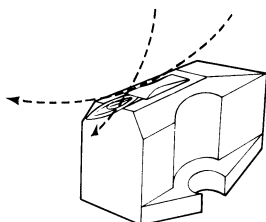
Method of handling record without touching recorded area.

How to Clean the Stylus

MOST cartridge manufacturers have provided little or no information regarding the proper methods of stylus cleaning. However, with the advent of ultra-lightweight tracking cartridges and increasing use of the elliptical stylus, this process has become as necessary as the cleaning of the record itself. The most important consideration is that the stylus must never be allowed to build up an accumulation of "goo" and pollution so that a coating is formed around the surface, riding in the grooves of the record. Therefore, the stylus should be inspected and cleaned after each playing session and immediately after playing any record which may have received applications of anti-static fluids.

If this procedure is used, no deposit will build up or harden and the stylus will remain perfectly clean to the end of its useful life. There are various types of stylus cleaning brushes and fluids available, some of which are more effective and useful than others. Testing the effectiveness of a cleaning tool is a simple matter of viewing the stylus under a microscope of X10 magnification. The choice of a cleaning fluid must be carefully checked, as some pickup cartridges can be damaged by the chemical reaction. In general, frequent dry cleaning by the use of a brush with short bristles or the WATTS Stylus Cleaner is the safest course to follow.

When cleaning the stylus, the pickup cartridge is held securely and the brush or pad is stroked across the stylus in a forward motion. In the case of an elliptical stylus, the cleaner should be stroked across the stylus in a 45 degree angle, from right to left, and left to right.



How to Play Records with Record Changers

FROM their delicate nature, it becomes apparent that records were originally designed and developed to be played singly and not piled one on top of the other. However, when we consider the convenience (not to mention the commercial aspects of record production) we recognize the necessity for the development of the "automatic record player" or record changer.

While it is true that most, if not all, of the older record changers posed some serious problems of distortion and varying stylus pressures and tracking angles, some of today's better units have in a large measure remedied these problems. Nevertheless, we are still faced with the problem of groove damage caused by one record dropping on another while the turntable is rotating. This damage is vastly increased by the abrasive action of dirt, dust and grit, which may be lodged in the grooves of either or both records.

It is essential to reduce this "wear and tear" to a minimum. The obvious solution is to remove all possible dirt, dust and grit before stacking. This is best accomplished by properly cleaning each record (*on both sides*) before playing. Fortunately, slitting the sleeve as previously described facilitates the cleaning process. For all we need do now is to place the record in its sleeve on a clean surface, open the sleeve, clean one side with either the Preener, Manual Parastat or Hi Fi Parastat, turn it over and clean the other side and then place it on the Record Changer spindle. In this manner we minimize the problem of record changer stacking.

How to Store Records

IDEALLY, records should be stored in a closed cabinet, where the humidity can be controlled. If a sufficient degree of humidity is maintained in the cabinet, we will be able to avoid some of more harmful aspects of static electricity. The installation of some form of atmospheric humidifier which can maintain the moisture in the cabinet at a 75/80% relative humidity will maintain records which have been inerted of static charge free of static.

Additionally, records should be arranged vertically, with the open ends of the jackets toward the back of the cabinet. Further, any open space between the records should be filled with books, or similar objects so that the records are kept upright under gentle pressure. In this way we prevent warpage.

Lastly, records should never be stored where excess temperatures can result. Storage of records in direct sunlight can easily warp records.

Remember, records like people, function best under moderate temperature and humidity conditions.

Conclusion

THE proper care and maintenance of records is really a commonsense procedure.

Once we understand and appreciate those conditions or circumstances which contribute to record wear and deterioration, and which create static electricity, we are in much better position to do something about it.

Since a combination of static electricity, dust, dirt and other pollutants are the arch enemies of proper record care, we must do everything we can to eliminate or minimize their effects.

The following particular points are worthy of repetition and re-emphasis:

1) On no account must the same implements be used treating new records as have been previously used on old records which were treated with cloths or sprays.

2) The Preener is designed for the treatment of new, or untreated records *only*. The Manual Parastat is intended for use with discs which have received previous anti-static applications, but a second separate one may be used instead of a Preener for new records. The Hi Fi Parastat is intended for use with new or like-new clean records to remove the last traces of micro-dust to enable the ultra-lightweight pick-up cartridge with and without elliptical stylus to track properly.

3) The cleaning device *must* be thoroughly cleaned and reactivated before application.

4) The turntable mat should be wiped clean before each playing session. A used Preener is ideal for this purpose.

5) The pick-up stylus should be examined with a X 10 magnifier, and cleaned after each playing session. *The state of the stylus gives a clear indication of the state of the records.*

6) During the application of *any* cleaning device, it is of the utmost importance:

(a) to see that it is not more than *barely damp or humid*.

(b) that the record is revolved *very slowly*. If it is allowed to rotate too quickly, cleaning will be inefficient and static is likely to be generated.

7) The new Formula Fluid *must not* be used at lower dilutions than those recommended in the instructions which accompany our various cleaning tools.

8) Atmospheric humidity is crucial in the control of static. The worst troubles are invariably found in artificially dry conditions where windows are opened little or seldom. In such circumstances, some form of atmospheric humidifier is desirable.

With the proper observance of the practices outlined in this booklet, records should last indefinitely and remain a continuing source of pleasure and gratification.